DATA ANALYTICS CAPSTONE

Enterprise Data Reconciliation

Turning three messy systems into one CFO-ready revenue dashboard

Python · SQL · Regex · pandas

Three systems, none of them clean

47,464

HR database (SQL)

Rows of historical status changes for 25,000 users. Most users appear more than once.

100,000

Server logs (text)

Raw log lines. The transaction data is buried inside text strings, mixed with error lines.

104 of 365

Exchange rates (CSV)

Days with no EUR to USD rate. Markets are closed at weekends, so those dates are blank.

Goal: a Q1 to Q4 revenue dashboard in USD, built from Active customers only.

The pipeline in four steps

STEP 1

SQL

Newest status per user, then keep Active only.

14,183

active users

STEP 2

Regex

Pull date, user, product and euros out of the logs. Drop errors.

94,902

transactions

STEP 3

Merge

Forward-fill weekend rates, then join logs, rates and users.

53,899

rows in the final table

STEP 4

Chart

USD revenue is euros times the daily rate. Group by month and by customer.

$148.3M

total USD revenue

End to end in about a second: vectorised pandas, no loops.

Trap 1: users appear more than once

```
SELECT user_id, name, status, updated_at
FROM (
  SELECT *, ROW_NUMBER() OVER (
    PARTITION BY user_id
    ORDER BY updated_at DESC) AS rn
  FROM users
)
WHERE rn = 1 AND status = 'Active';
```

47,464
rows in the status log

25,000
unique users

14,183
whose newest status is Active

Order matters: take each user's newest row first, then filter to Active.

Trap 2: transaction data hidden in text

```
[2023-03-15 00:00:00] INFO: Transaction processed.
payload={"u":"U-19484", "item_code":"PRD-874", "eur_val":3404.44}
```

↓ str.extract() with one regular expression

Date	User_ID	Product_ID	Euro_Value
2023-03-15	U-19484	PRD-874	3404.44

100,000 log lines – **5,098** ERROR lines = **94,902** transactions

Trap 3: weekends have no exchange rate

PLAIN INNER JOIN

67,918

rows kept. **28.4%** of transactions silently dropped, because their date has no rate.

FORWARD-FILL (FFILL)

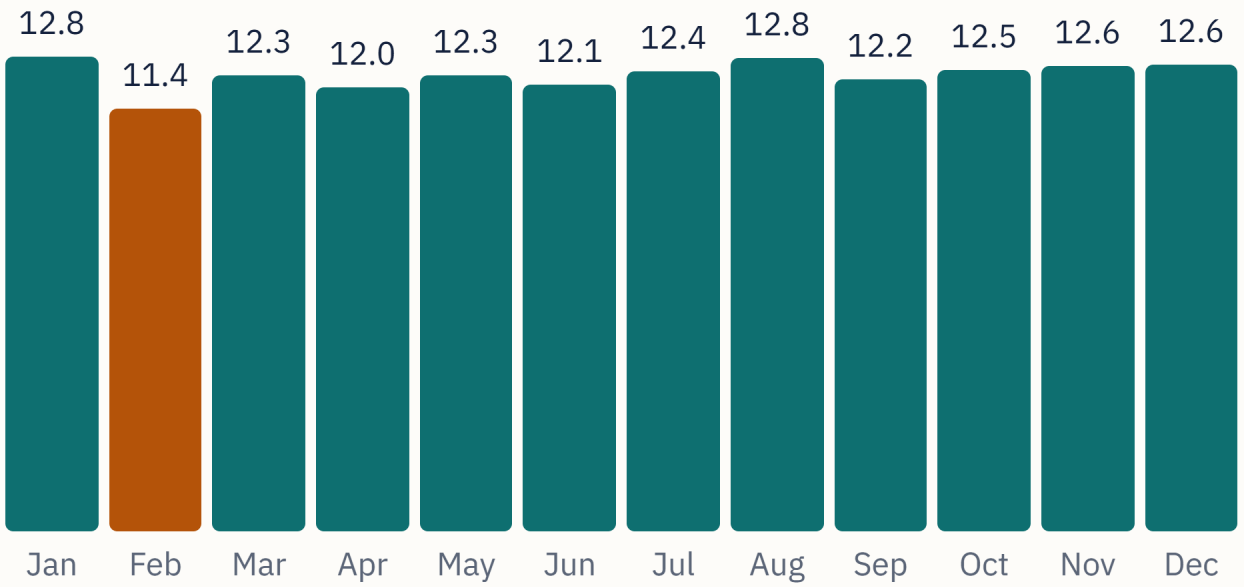
94,902

rows kept. Friday's rate rolls into Saturday and Sunday, so **nothing is lost**.

Of 365 days, 104 have no rate. Each transaction is matched to a rate by its exact date.

Dashboard: steady revenue, a tight top five

Total USD revenue by month ($M)



About $12.4M a month. February is lowest at $11.4M.

Top 5 customers by USD spent

U-06582	$40,051
U-23313	$39,556
U-03805	$39,530
U-20018	$38,650
U-06489	$38,629

https://github.com/surajmalik430git/enterprise-data/blob/main/README.md

$148.3M
total USD revenue from Active customers

$12.4M
average a month. Low: Feb $11.4M. High: Jan $12.8M

U-06582
top customer, about $40,051 spent

Built with Python · pandas · SQLite · regex · matplotlib · seaborn

Runs on 100,000 log lines in about a second, with no loops.

GitHub: [add your repository link]